Apache Airflow

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Apache Airflow

- A framework for building data engineering pipelines in python
- Created by Airbnb, now an open source Apache Software Foundation Project
- Purpose is to automate and orchestrate complex data pipeline and other workflows
- Allows programmatic authoring, scheduling, and monitoring of workflows as DAGs
- Contains a web server, a scheduler, a metastore, a queueing system, and executors
- Can run as a single instance or can run on a cluster with many executor nodes
- Uses DAGs where each node is a task

When to use ?

- When workflow has clear start and end and run on a schedule
- Prefers python coding to build workflow
    - Provides version control, team collaboration, testing and extendibility
- When pipeline is complex and recurring
- Want to process historical data, rerun failed tasks
- Designed for batches workflows

Airflow: Architecture

- **Web Server (UI):** Provides a visual interface to manage and monitor workflows.
- **Scheduler:** Monitors DAGs and triggers tasks based on time or events.
- **Executor:** Manages how tasks are run, distributing them to workers (e.g., SequentialExecutor, CeleryExecutor, KubernetesExecutor).
- **Workers:** Execute the tasks.
- **Metadata Database:** Stores DAGs, task states, and other metadata.

Apache Airflow: DAGs key attributes

- Schedule : when workflow should run
- Tasks : discrete units of work that are run on workers
- Task dependencies : the order and conditions under which tasks execute
- Callbacks: actions to take when entire workflows completes
- Additional Parameters: many other operational details

Use Cases

- **ETL (Extract, Transform, Load) pipelines:** Manage and automate data ingestion, transformation, and loading.
- **Machine learning workflows:** Orchestrate training, evaluation, and deployment of machine learning models.
- **Data warehousing:** Build and manage data pipelines for data lakes and warehouses.
- **Ad-hoc job scheduling:** Replace cron with more robust and flexible scheduling.
- Promotes code reusability and collaboration.
- Scalable and fault-tolerant.
- Provides visibility and control over data pipelines.

Apache Airflow Demo